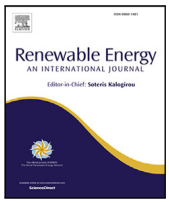
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Swin transformer-based transferable PV forecasting for new PV sites with insufficient PV generation data

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ABSTRACT

To effectively operate solar farms, accurate photovoltaic (PV) generation forecasting is required. For a newly constructed solar farm (PV site), its generation data could be limited. Using the sky images obtained from ground-based whole-sky cameras, this paper proposes a transferable Double route Shifted window Cross-Attention Transformer (DSCAT) framework to provide PV forecasting of the newly constructed PV site. The framework is trained using the data of an established PV site and then provides ultra-short-term PV forecasting for the newly constructed PV site. In the proposed framework, a temporal difference parallel Shifted window (Swin) Transformer-based structure is designed to capture the cloud motion details and extract the static spatial features. Then, a cross-attention structure is utilized to analyze the temporal features and predict the future PV generation. A variety of transfer strategies are designed to transfer the trained model to provide the PV forecasting at the new PV site. The training and transfer experiments are conducted with real-world sky images and PV generation datasets. The result shows the proposed framework could be transferred between varied environments, and provide a reliable forecast which achieves a 49% enhancement over the persistence baseline and 13% improvement over the PV forecasting benchmarks on average.

1. Introduction

Solar generation has become a considerable part of the power system [1,2]. PV generation facilities can be installed in various capacities, including residential, building, and solar farm levels [3]. Such a wide range of PV installations can lead to significant grid uncertainties. This requires reliable PV forecasting [4].

For real-time dispatch and electricity price control, the minutes-ahead PV generation forecasting (e.g. ultra-short-term PV forecast) is required [5]. However, the intermittent nature of PV generation poses a significant challenge for developing an accurate ultra-short-term PV forecast. The local weather conditions, the flowing and deforming clouds and the switching between direct sunlight and shadow can cause rapid variations in the PV generation [6]. To address the challenge, numerous numeric series-based PV forecasts and sky-image-based PV forecasting methods have been developed.

1.1. Numeric series-based photovoltaic forecasting

In some studies, the numerical weather prediction (NWP) and historical PV generation data are combined to feed a hybrid deep learning model for PV generation prediction as in [7]. In [8], the self-attention mechanism and the Long Short-Term Memory (LSTM) were considered to utilize the NWP data effectively. A wavelet decomposition was implemented with the convolutional neural network (CNN) and the LSTM models. In [9], the attention mechanism of the Transformer model was tailored to accommodate the spatio-temporal characteristics for PV forecasting. The PV generation from the neighboring PV sites was utilized to analyze cloud distribution and motion. Similarly, the architecture proposed in [10] modeled the PV generation of several PV sites to explore the spatial features. On the contrary, in [11] the topological distribution of a large PV fleet was analyzed, and subsequently the generation of the fleet was estimated through data collected from one PV unit i.e. the pilot unit.

In the above works, the local weather conditions and cloud motion for the target PV sites are inferred from the NWP and PV generation from the neighboring PV sites. Actually, satellites and whole sky cameras can provide continuous cloud images and reveal the static and dynamic stats of clouds [12,13]. Numerous algorithms have been proposed to forecast PV generation using sky images.

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1.2. Whole sky image-based photovoltaic forecasting

In [14], a CNN-based structure was adopted, in which the sky images were stacked and fed into a residual CNN to extract the spatio-temporal features. An additional parallel analysis of exogenous data was implemented for obtaining the temporal features. To further explore the hidden features embedded in the sky images, a combined Vision Transformer (ViT) and Gated Recurrent Unit (GRU) framework was proposed in [15]. Differently, in [16] two paralleled spatial and temporal data streams were constructed. The original RGB sky image and optical flow map were fed into a CNN and a separate convolutional long short-term memory (ConvLSTM) model to extract spatial and temporal features. In [17] the Cuboid descriptor and optical flow were combined for cloud dynamic feature extraction while a dense convolutional network was used for the cloud static feature extraction. The above methods achieved higher accuracy compared to several baseline spatio-temporal PV forecasting frameworks. This proves that the utilization of image processing techniques can enhance the representation of the cloud's dynamic features. In [18], a mathematical cloud speed calculation method was combined with a neural network to infer future sky images. Then, the weather classification based on the sky images was conducted leading to different mapping models. However, the camera used in this study kept the sun at the center of the image. This limits its applicability and requires additional preprocessing. To analyze detailed seasonal and trend information from the historical PV generation data, a sky-image-derived decomposition method was utilized in [19]. The encoded feature maps derived from the sky image sequence were fed into the attention block to guide the decomposition of the PV generation data. Moreover, a clustering algorithm was implemented to facilitate the model recognizing different scenarios. The model outperformed several baseline models. However, the clustering algorithms require a long-term, informative dataset to capture season features. Also, in the ultra-short-term PV forecasting, the seasonal feature was not significant.

1.3. Transfer learning-based forecasting strategy

For a new PV site, its historical PV generation data and sky images may not be sufficient for producing accurate PV forecasting. Transfer learning, which aims to enhance the performance of deep learning models on small datasets, is suitable for solving this problem [20]. Several transfer-learning-based PV forecast strategies are proposed in the literature.

Shallow transferring and deep transferring were both implemented in [21]. The utilized Constrained Long Short-Term Memory (CLSTM) model was frozen in six different layer combinations and evaluated. However, the method proposed in [21] was only for day-ahead forecasting. The numeric-only inputs (i.e. PV generation) are not sufficient for ultra-short-term forecasting. A variant of the LSTM model, the Attention-Dilated CNN-Bidirectional LSTM Network (ADBN), was applied with transferring strategies in [22] to provide forecasting of the new PV sites with insufficient datasets. The authors in [22] also deployed five different parameter transfer strategies to find the optimal strategy. The result shows that the transfer strategies could improve the accuracy and reduce the training time. Still, only the numeric weather information and PV generation records with 5-minute intervals are fed. This cannot provide comprehensive details. Moreover, in [23] the authors investigated the transfer strategies on several renewable energy forecasting tasks. In the investigations, only the numeric PV and weather data were utilized. In [24], the transfer strategies were utilized to enhance the performance of the graph neural network on data-scarce PV scenarios. However, only the numeric PV generation data was used. As an informative data source, the sky image should be considered as the essential part of the dataset to help apply the transferred ultra-short-term PV forecasting models to the newly built PV site which has limited PV data. The transferring of the sky image-based PV forecasting model was discussed in [25]. The result shows the feasibility of transfer learning strategies for the sky image-based ViT-backbone model. However, further performance improvement is still required.

1.4. Contributions

Considering the aforementioned limitations of the existing methods, there is a lack of a generalized ultra-short-term PV forecasting framework based on sky image data sources. Also, the role of sky images in helping the PV forecasting task on newly built PV sites (with insufficient data) is not well-focused. Therefore, the key contributions of this study are as follows:

1. A generalized framework, the Double route Shifted window Cross-Attention Transformer (DSCAT) is proposed for PV generation forecasting based on sky images. The Swin Transformer backbone is utilized to capture both the local and global spatial features and improve the performance of ultra-short-term PV forecasting. Moreover, the modular design of DSCAT provides the capability of transferring to new datasets.
2. A double route and cross-attention structure is proposed for analyzing the inner spatio-temporal relation of the sky image sequence. The pairing of spatial and temporal features precisely maps the historical cloud motion to future cloud distribution, which indicates the future PV generation situation.
3. The detailed transfer learning strategies are designed and evaluated. The proposed strategies provide a novel and time-efficient method for PV forecasting at newly constructed PV sites with sky images.

The remainder of this paper is organized as follows. Sections 2 and 3 elaborate on the problem formulation and the methodology used. Section 4 illustrates the settings of experiments. The evaluation and discussion of results are presented in Section 5. Finally, the paper is concluded in Section 6.

2. Problem formulation

In this paper, the forecasting model is tasked to predict the future PV generation for a newly constructed PV site, which has limited data (PV generation records and/or sky images). The dataset is referred to as the target domain data. To train the model, the data from well-established PV sites (having sufficient PV generation data and sky images) are used and it is referred to as the source domain data.

The model is initially trained on the source dataset D^s and then transferred to the target dataset D^{tg}. In the source domain, the continuous n sky images $I_t^s = (i_{t-n}^s, \dots, i_{t-2}^s, i_{t-1}^s)$ with historical timestamps are fed as the main spatio-temporal inputs, in which t denotes the current timestamp. The following data from the source domain are also utilized in the training: (1) PV generation G_t^s; (2) the specific timestamp R_t^s (represented as the ratios between hours to day and day to year from 0 to 1. E.g. 3 AM is represented as 0.125, and 31st December is represented as 1); and (3) estimated clear-sky PV generation E_t^s. For the time horizon Δ, its PV generation of the source domain $\hat{P}_{t+\Delta}^s = (\hat{p}_{t+\Delta}^s, \hat{p}_{t+\Delta+1}^s, \dots, \hat{p}_{t+\Delta+n-1}^s)$ is predicted. Then, within the proposed nonlinear forecasting model M^s, the source domain forecasting could be denoted as (1).

$$\hat{P}_{t+\Delta}^s = M^s(I_t^s, G_t^s, R_t^s, E_t^s) \tag{1}$$

After the training using the source domain dataset, the parameters from M^s will be partially transferred to the target domain dataset D^{tg} to provide a forecast of the PV generation at the target PV site. The transfer learning stage could be denoted as (2).

$$M^{tg} = T(M^s, D^{tg}) \tag{2}$$

where T denotes the transfer strategies.

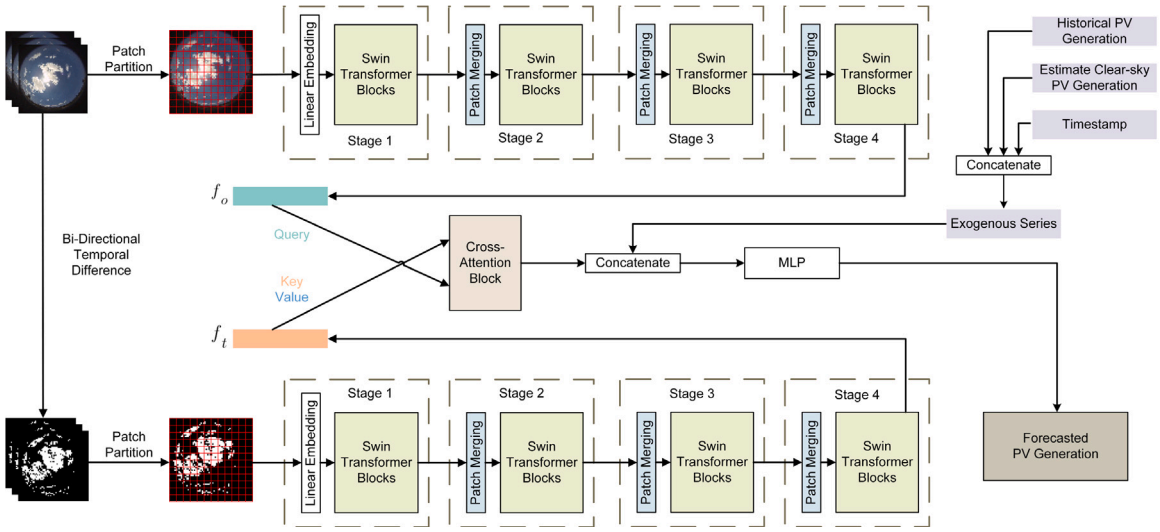


Fig. 1. Structure of the DSCAT framework.

3. Methodology

The structure of the proposed double route shifted window cross-attention transformer (DSCAT) framework is shown in Fig. 1. The sky image sequence I_t is the main input fed into the framework. A temporal difference algorithm is implemented on the original sky image sequence to obtain the dynamic pixel difference sequence. Both the original sky image sequence and the temporal difference sequence will be fed into the same Swin Transformer model to extract the hidden spatial features. After that, a cross-attention Transformer is allocated to catch the temporal information from the above two spatial feature sequences. The estimated clear-sky PV generation E_t, the ratio timestamp R_t, and the historical PV generation difference G_t^{diff} are all concatenated as the exogenous series to facilitate defining an approximate PV prediction range for the model. Finally, the future PV generation $\hat{P}_{t+\Delta}$ is predicted by the forecast head, which is a multilayer perceptron (MLP).

3.1. Double route spatial feature extraction

Most of the local weather conditions including cloud distribution, solar irradiance, wind speed and direction are the main effective factors in the variation of PV generation. To accurately predict future PV generation, it is necessary to extract these essential spatial features from the sky images.

3.1.1. Bi-directional temporal difference

Temporal difference, a moving objects segmentation algorithm, can extract the moving object from a relatively static background. The direct difference between two adjacent images ($Image_2 - Image_1$) is used to represent the moving trajectory of the object. In a sky image dataset with 1-minute resolution frames, the differences in clouds and the sun are clearly detectable and the clear sky background remains relatively unchanged. However, there could be various directions of cloud motion between two frames, which could be ignored by the original temporal difference algorithm. Thus, an improved bi-directional temporal difference algorithm is implemented to capture the cloud variations in both temporal directions as shown in Algorithm 1. In the algorithm, d denotes the difference matrix of two adjacent frames, d_{Neg} and d_{Pos} denote the difference matrix's negative and positive parts respectively, d_{Abs} denote the absolute value of the difference matrix.

After applying the above bi-directional temporal difference algorithm, the cloud motion in both time directions is extracted and fed into the Transformer module with the original sky images kept in parallel. This pre-extracted cloud motion feature helps maintain the clarity of temporal features in the vision of the Swin Transformer model.

Algorithm 1 Bi-Directional Temporal Difference

for Every two continuous frames in the input series **do**
  $Frame_1 \leftarrow grayscale(Frame_1)$
  $Frame_2 \leftarrow grayscale(Frame_2)$
  $d \leftarrow Frame_2 - Frame_1$
  if where $d < 0$ **then**
    $d_{Neg} \leftarrow d$
    $d_{Neg} \leftarrow |d_{Neg}|$
  else if where $d \geq 0$ **then**
    $d_{Pos} \leftarrow d$
  end if
  $d_{Abs} \leftarrow |d|$
  for $d_{Neg}, d_{Pos}, d_{Abs}$ **do**
    if $element < threshold$ **then**
      $element \leftarrow 0$
    end if
  end for
  return $d_{Neg}, d_{Pos}, d_{Abs}$
end for

3.1.2. Swin transformer block

Swin Transformer (ST), which is a variant of ViT, is proposed in [26]. This model has combined both the benefits of the CNN-based models and the vanilla Transformer model. Compared to classical CNN-based models, the attention mechanism incorporated in the Transformer can provide broader global vision information across the entire image. Moreover, the shift-window mechanism enables the model to extract detailed local features for small objects [27]. Based on the characteristics of the sky image-based PV forecasting task, the local features are required for cloud segmentation and localization of the spatial relations of the sun and nearby clouds. Also, the global features are essential for the comprehensive cloud distribution analysis and the potential effects of marginal clouds.

Scaled Dot-Product Attention, which is the core of the Transformer backbone [28] is used as the feature extractor. With the trained matrices query Q and the key–value pairs (K-V), the weighted output is generated based on the latent relationship between Q and K-V. The calculation between the query and key produces the weight for the corresponding value matrix, after which the weighted value is generated as the extracted features. The calculation could be denoted

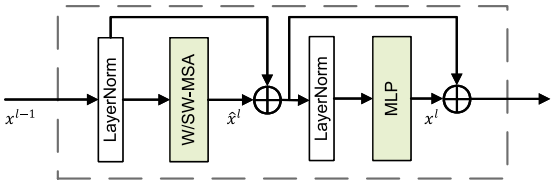


Fig. 2. Structure of the Swin Transformer block, where W-MSA is Windows Multi-head Self-Attention and SW-MSA is Shifted Windows Multi-head Self-Attention.

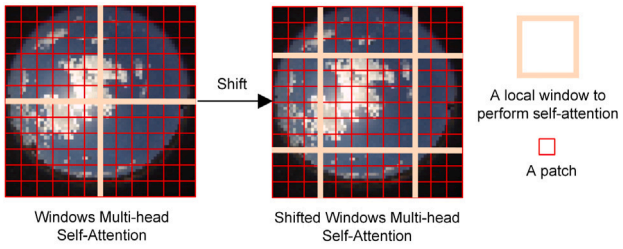


Fig. 3. Window shifting of two consecutive blocks.

as (3).

$$Attention(Q,K,V) = softmax(\frac{QK^T}{\sqrt{d_k}})V \tag{3}$$

The product of QK^T is divided by the dimension of key d_k to stabilize the gradient. Then, the self-attention mechanism is utilized. The query, key, and value matrices are all generated from the same input, which could extract the hidden features based on the internal correlation of the input. Moreover, multiple QKV matrices are trained to capture different aspects of one input (i.e. multi-head attention).

In the Swin-Transformer-based model, the patch partition step splits the input images into small patches. Then, the patches will be linearly embedded to feature vectors from size $H_p \times W_p \times C$ to d_f, where (H_p, W_p, C) denote the height, width, and channel of the patch, and d_f is the feature dimension. This step maps the 3-dimensional images to 1-dimensional feature vectors and feeds them into the ST blocks as Fig. 1 shows. After the first embedding, the size of the input is changed from $H \times W \times C$ to $\frac{H}{H_p} \times \frac{W}{W_p} \times d_f$.

The structure of the ST block is shown in Fig. 2, where the LayerNorm means the layer normalization (LN) operation. The two successive ST blocks with both window and shifted window (SW) multi-head self-attention (MSA) can be formulated as (4).

$$\begin{aligned}
\hat{x}^l &= W\text{-}MSA(LN(x^{l-1})) + x^{l-1} \\
x^l &= MLP(LN(\hat{x}^l)) + \hat{x}^l \\
\hat{x}^{l+1} &= SW\text{-}MSA(LN(x^l)) + x^l \\
x^{l+1} &= MLP(LN(\hat{x}^{l+1})) + \hat{x}^{l+1}
\end{aligned} \tag{4}$$

where $\hat{x}^l$ and x^l denote the output of the W-MSA/SW-MSA and MLP modules of layer l respectively.

Unlike the ViT which implements self-attention across all the patches to capture the global vision, the ST first utilizes MSA in the regularly partitioned windows to extract local features. In the next consecutive block, the windows shift across the image to provide the information exchange between adjacent windows as shown in Fig. 3.

In the proposed framework (Fig. 1), there are four consecutive stages within the Swin-based Transformer block. Before stages 2, 3, and 4, there are three patch merging modules. The neighboring 2×2 patches will be merged into one patch and compressed by a linear layer, the output feature dimension will become twice the input feature dimension. For example, the patch merging of stage 2 will merge the input from $\frac{H}{H_p} \times \frac{W}{W_p} \times d_f$ to $\frac{H}{2H_p} \times \frac{W}{2W_p} \times 2d_f$. Therefore, the windows in deep layers contain the features of multiple windows in shallow layers, the ST model could gradually capture the global vision with the same window size.

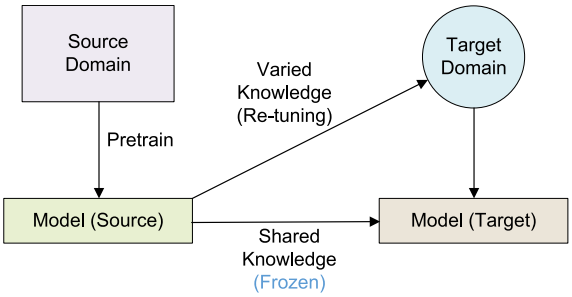


Fig. 4. Transfer learning between knowledge domains.

3.2. Cross-attention temporal feature extraction

After the double route spatial feature extraction, two spatial features from the original image sequence and temporal difference sequence are generated. As the PV forecasting task is concerned with future information, the hidden spatial and temporal features need to be combined and mapped to the future spatial features. Therefore, the cross-attention mechanism is utilized over the original spatial features f_o and temporal difference f_t features as a decoder to generate future spatial features. In this block, the f_o is fed as the query input called target sequence, and the f_t is treated as the key and value input called source sequence. The corresponding information will be retrieved from f_t based on the positional encoding and spatial features of f_o, which represents the hidden future spatial features. This block could be formulated as:

$$Cross\text{-}Attention(f_o, f_t) = Attention(Q_{f_o}, K_{f_t}, V_{f_t}) \tag{5}$$

where $Attention$ is presented in (3).

After the cross-attention block, the hidden spatio-temporal features are combined with the exogenous series: (1) historical PV generation; (2) the time of day and year; and (3) estimate clear-sky PV generation. Finally, an MLP is implemented to generate the predicted future PV generation.

3.3. Transfer strategies

A well-performed neural network could be obtained with sufficient information from a large and informative dataset. However, it is challenging to train a neural network on a new and small dataset, such as the data from a newly built PV site. Therefore, transfer learning, which is designed to transfer knowledge from the source domain to the related target domain, is appropriate for solving the forecasting problem on newly built PV sites. The base structure of transfer learning is shown in Fig. 4.

In this paper, the source domain dataset and the target domain dataset may be different because of the different PV panel parameters, locations, and weather patterns of the established (old) PV sites and the newly built PV sites. However, the structures of the datasets across domains are similar, and the tasks of mapping historical data to future PV generation are the same. Therefore, the neural networks are able to learn the knowledge from the established large dataset and transfer it to the new small dataset.

There are several transfer learning strategies T that can be implemented in deep learning models. By freezing various stages of parameters in the entire forecasting structure, the model can retain the corresponding knowledge learned from the source domain. Then, the other parameters will be fine-tuned on the target domain to learn the distinct pattern from the small target domain data. In this paper, it is necessary to keep the generic knowledge between the two PV sky

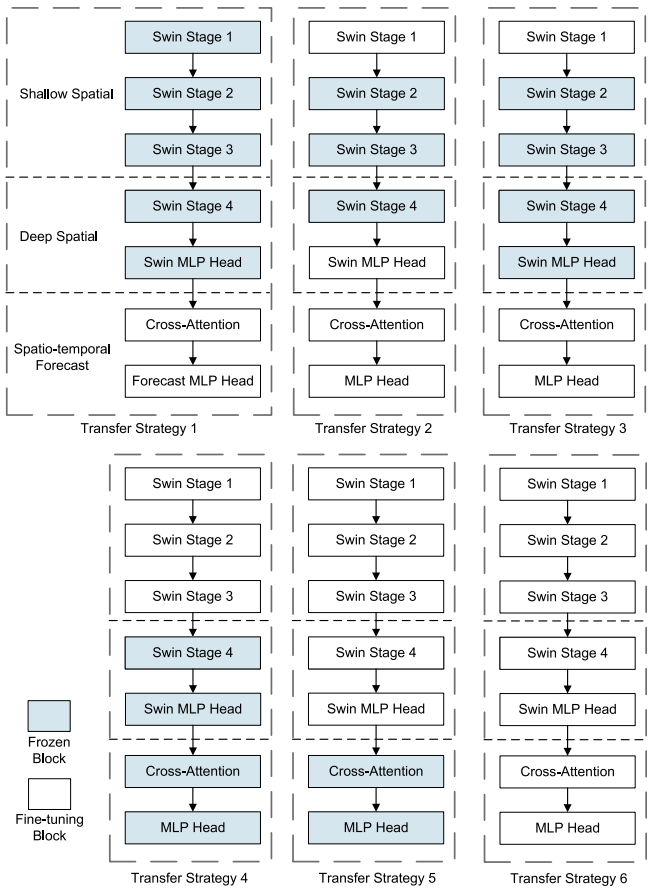


Fig. 5. Transfer strategies on PV forecasting module.

image datasets and retrain the remaining parts to fit the target dataset. With the clearly hierarchical model structure in this paper, the transfer strategies are designed based on various freeze areas.

Fig. 5 lists the implemented strategies in this research. The parameters of the frozen blocks kept the same as the source model M^s and the parameters of the fine-tuning blocks have been re-trained on the target dataset. The forecasting framework is roughly split into three parts: shallow spatial module, deep spatial module, and spatio-temporal forecast module. Based on the various aspects of knowledge to be remained, six transfer strategies have been designed. For freezing different levels of the spatial ST module, Strategies 1 to 3 are tested. Also, Strategies 4 and 5 are tested to verify the similarity of the across-attention spatio-temporal features between source and target domains. Finally, Strategy 6 fine-tunes all the layers to compare the effect of the retained knowledge from the source domain. After the transfer stage, the retained layers and fine-tuned layers are combined as the target model M^{tg}.

4. Dataset and experiments

4.1. Datasets

Two sky image datasets with PV generation are utilized for source training and transfer learning. The source PV dataset is of a 30.1 kW PV panel allocated at the Stanford University campus. The corresponding sky image dataset is collected by a whole sky camera 125 m away from the PV panel arrays [29]. The PV generation data is recorded in a 1-minute resolution and is minutely averaged. The original sky videos are recorded at 20 frames per second with 2048 × 2048 pixels and RGB channels. To reduce the unexpected computational redundancy, the sky

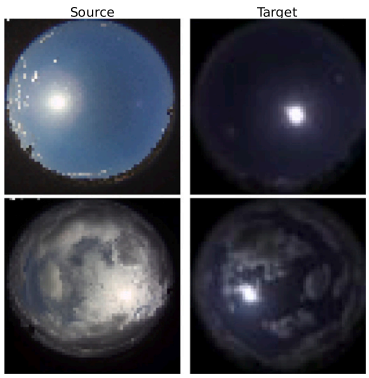


Fig. 6. Sample sky images (Left column: Stanford University dataset. Right column: the University of Queensland dataset).

images are extracted from the original video at a 1-minute interval, and then they were resized to 64 × 64. The time period of the source dataset is from March 2017 to December 2019.

As the target domain, the other sky image dataset is collected from a hemispheric camera in a solar farm at The University of Queensland's Gatton campus, which has a different peak capacity, environment, and weather pattern with respect to the Stanford University campus. The original target sky images have 2408 × 1536 pixels and RGB channels with 10 s resolution. The target PV generation dataset has the same time resolution as the source dataset which is 1 min. The target dataset was collected from January 2020 to May 2020. The original UQ Gatton dataset has been published at [30]. The pre-processing was implemented on the target dataset to match the parameters of the source dataset. The samples of sky images from both the source and target datasets are shown in Fig. 6.

There are outlier data points caused by the record device such as unchanged PV generation for a long period and zero PV generation in both the source and target datasets. All outlier data points are deleted to guarantee the quality of PV generation data.

As the exogenous data, the estimated clear sky PV generation data for both the source and target domains are generated from the Ineichen and Perez clear sky model [31] using the corresponding location and PV panel parameters.

4.2. Pre-processing

4.2.1. Source dataset

With the pre-selected sky images in the dataset, the image size does not need to be modified. The RGB values of images were scaled from [0, 255] to [0, 1] to avoid the gradient vanishing. Also, the PV generation data G was min–max scaled to [0, 1] by (6).

$$G_{minmax} = \frac{G - G_{min}}{G_{max} - G_{min}} \tag{6}$$

where G_{min} and G_{max} are the minimum and maximum values of the entire PV generation series. Additionally, some records with unchanged values in a period are removed.

4.2.2. Target dataset

To ensure compatibility with the generally available input size across various types of sky cameras, the images in the target dataset are cropped to 1500 × 1500. This is to remove non-informative margins in the original sky images. Then the cropped images are resized to 64 × 64 to match the source dataset. Additionally, the same scaling algorithms on images and PV generation data are also implemented on the target dataset as on the source dataset. The sampling resolution of the target dataset is set to one minute.

Table 1
Hyper-parameters of training.

Hyper-parameter	Configuration
Batch size	8
Learning rate	0.0001
Epochs	40
Length of inputs (mins)	5
Length of output (mins)	5
Forecasting horizon (mins)	5
Input/output time resolution (min)	1

Table 2
Structural parameters of DSCAT.

Parameter	Configuration
Swin Transformer Module	
Hidden feature dimension	96
Number of layers (per stage)	(2, 2, 6, 2)
Number of heads (per stage)	(3, 6, 12, 24)
Downscaling factors (per stage)	(4, 2, 2, 2)
Head dimension	16
Windows size	2
Spatial feature dimension	256
Position embedding	Relative
Cross-Attention module	
Hidden feature dimension	256
Number of heads	4
Head dimension	64
Forecasting MLP Head	
Input Layer	(4 × 256 + 11, 512)
Hidden Layer	(512, 256)
Output Layer	(256, 5)

4.3. Experiment setting

All the experiments are conducted on high-performance computing (HPC) GPU nodes. Each model training task is implemented on an individual HPC node (having two Intel Xeon Gold 6132 28 cores CPUs, four NVIDIA SXM2 T 32 GB V100 Accelerator Units, and 384 GB of DDR4 RAM). The proposed DSCAT and other several baseline models are all developed using Python 3.8.11 and PyTorch 1.9.1.

Both the source and target datasets are split to train, validation, and test sets. Moreover, the test sets are manually chosen to cover most possible weather conditions. In the source dataset, the test set consists of ten sunny days and ten cloudy days chosen. Following a similar dataset size ratio, three sunny days and five cloudy days in the target dataset are selected for testing. Any data in the days of the test set will not occur in the train and validation sets. Then, the rest of the records are randomly split into a ratio of 80%/20% as the train and validation sets.

After several preliminary tests with a small pre-validation dataset, the hyper-parameters are decided to optimize the model's convergence speed while minimizing unnecessary computational burden. At the training stage, the training epochs are set to 40 to ensure the quality of the models. Moreover, to avoid overfitting on the training sets, the training process is stopped with four continuous epoch error increases on the validation dataset. There is no dropout set with the experiment to prevent the missing essential parameters for the regression task. All the hyper-parameters are listed in Table 1.

Also, to save computational resources and enable quick deployment to reach the transfer target, the number of Transformer layers in all the blocks is chosen to be small. Moreover, to ensure the same lengths of the double route sequence, the extracted feature of the furthest-in-time sky image was ignored. The parameters of the DSCAT framework are shown in Table 2.

Additionally, the random seed is set to 0 across all the experiments to ensure the same dataset split and model parameter initialization.

4.4. Baseline methods and metrics

To evaluate the performance of the DSCAT framework, a baseline and various benchmarks are chosen for comparison. Firstly, the smart persistence (SP) [32] is used as the traditional mathematical baseline. This model assumes that the ratio between the real PV generation and the estimated clear-sky PV generation remains constant across the forecast horizon. Thus, the forecasting result $\hat{P}_{t+\Delta}^{Persist}$ of this model could be denoted as (7).

$$\hat{P}_{t+\Delta}^{Persist} = \frac{G_t}{E_t} \cdot E_{t+\Delta} \tag{7}$$

On the other hand, four deep learning models are selected as benchmarks to cover various structures of the sky image-oriented neural network. The CNN, CNN+LSTM (CNNLSTM), and ConvLSTM models are implemented according to [33]. They are the most widely used convolutional and recurrent neural network architectures. Moreover, the original ViT [34], which is the state-of-the-art computer vision model, is also implemented for comparison. This is to evaluate the performance improvement of the shifted window mechanism adopted in the proposed DSCAT. In the implemented ViT, the layers and hidden feature dimensions are set as the same as the DSCAT which is 12 and 256.

For the series regression task, four metrics are selected to evaluate: Mean Absolute Error (MAE), normalized Root Mean Square Error (nRMSE), coefficient of determination R squared (R^2), and Forecast Skill score (FS), which are denoted as (8)–(11).

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{8}$$

$$nRMSE = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}}{y_{max} - y_{min}} \tag{9}$$

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y}_i)^2} \tag{10}$$

$$FS = 1 - \frac{nRMSE}{nRMSE_{SP}} \tag{11}$$

where y_i, $\bar{y}_i$, $\hat{y}_i$ denote the ground truth, the mean of ground truth, and the prediction. With the PV generation data was min–max scaled to [0, 1] by (6), the $y_{max} - y_{min}$ in (9) equals 1. Finally, the smart persistence algorithm is selected as the baseline reference forecast to calculate the forecast skill.

5. Evaluation and discussion

5.1. Forecast performance on source domain

Firstly, the DSCAT model was compared with the baseline and benchmark models on the manually chosen test set of the source domain to evaluate its forecasting performance.

The metrics comparisons amongst the DSCAT model proposed in this paper and the baseline models are shown in Fig. 7. The metrics in the top two graphs are MAE and nRMSE. The lower error value indicates the better model performance. As indicated by both metrics, the DSCAT model has the lowest error. According to the MAE values, all benchmark deep learning-based models (CNN, CNNLSTM, ConvLSTM, and ViT) are outperformed by the smart persistence baseline. It implies that even with extra information, these benchmark deep learning-based models (especially the CNNLSTM) cannot properly follow the PV generation curve. Compared with the ConvLSTM (the best-performing deep learning-based model), the DSCAT model achieved a 3.6% decrease in the MAE. This shows that the DSCAT model can accurately utilize the input information to enhance the overall trend fitting of the PV generation. Unlike MAE, nRMSE is more sensitive to outliers. Therefore, the smart persistence algorithm, which lacks robustness to fluctuations, has a relatively worse value of the nRMSE. The DSCAT

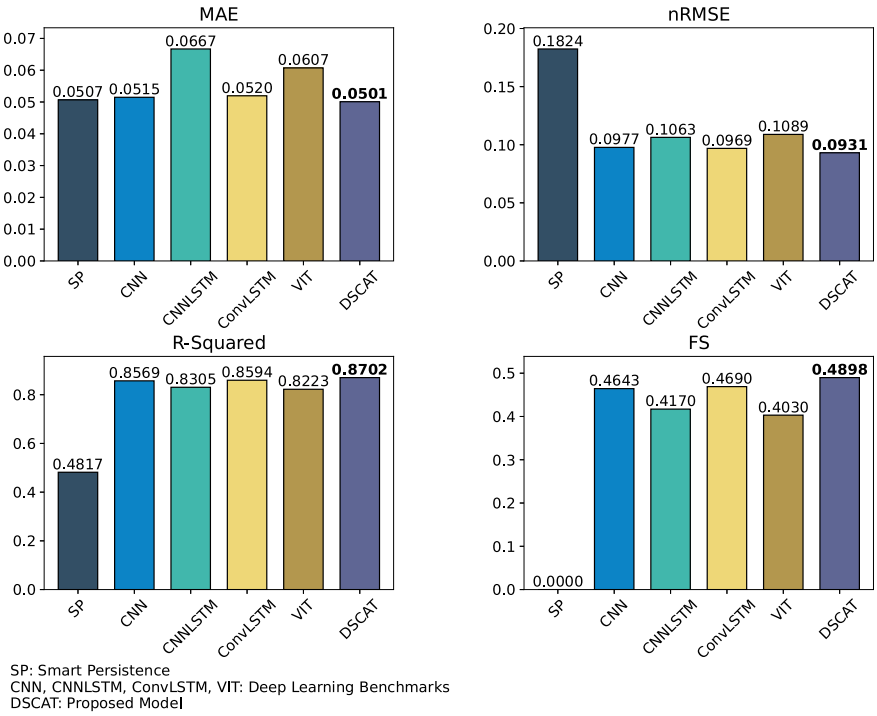


Fig. 7. Comparison of metrics on source test set.

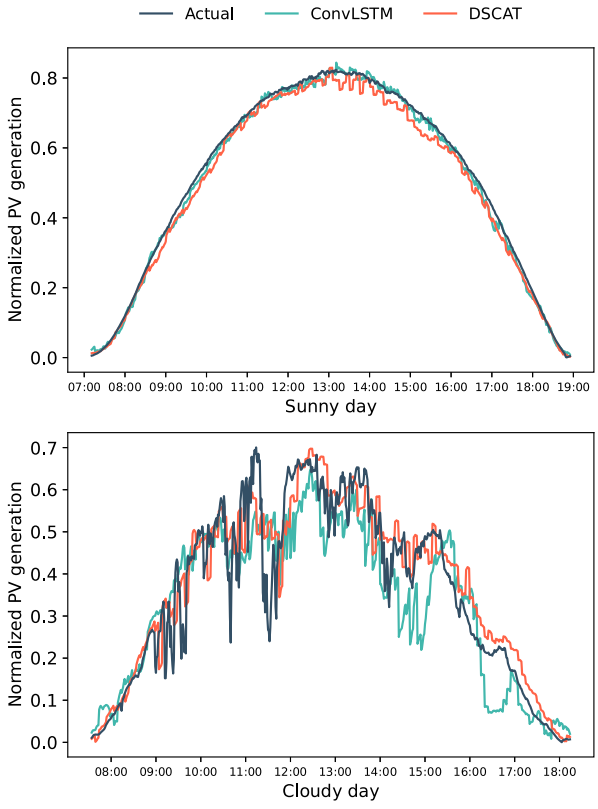


Fig. 8. The PV forecasting of one sunny day and one cloudy day in the source test set.

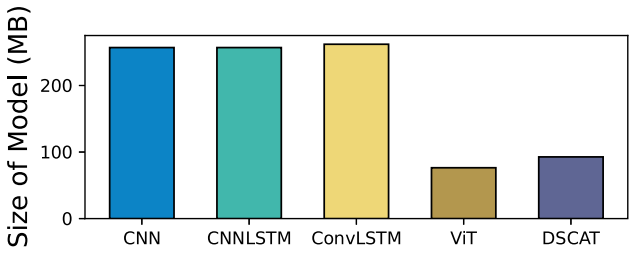


Fig. 9. The sizes of models.

model performs 48.98% better than the smart persistence algorithm. Among the benchmark deep learning-based models, the ViT performance is the worst (its nRMSE is 18.08% higher than that of the DSCAT model), while the three convolutional-based models achieve relatively good performance (they have lower nRMSE values when the spatio-temporal ConvLSTM structure is included). Overall, the DSCAT achieves the lowest nRMSE value, which is 3.92% lower than that of the best-performing benchmarks and 10.19% lower than the deep-learning benchmarks on average. This is because the spatial structure adopted in the DSCAT model considers both global and local features and combines double route temporal structure.

The bottom two graphs in Fig. 7 are R^2 and FS scores. The higher scores indicate better model performance. The R^2 score refers to the correlation between the change of input and output. The highest R^2 score of the DSCAT model reflects that it has the best capability to map the input trend to the possible future PV generation. Also, the DSCAT model attains the highest FS score, which is about 49% better than the smart persistence algorithm, and on average, it has a 13.41% FS score improvement over the benchmarks.

The PV forecasting results of the best-performed benchmark ConvLSTM and our DSCAT model are shown in Fig. 8. Both models achieve desirable performance on sunny days. Even though the ConvLSTM could roughly recognize the trends under on cloudy days, significant amplitude mismatches between the ConvLSTM forecasting result and the ground truth can be observed in the figure. This indicates that the ConvLSTM fails to capture the dynamics of cloud motion accurately. However, as shown in the figure, the DSCAT model proposed in this paper can precisely catch both trends and amplitude of PV generation.

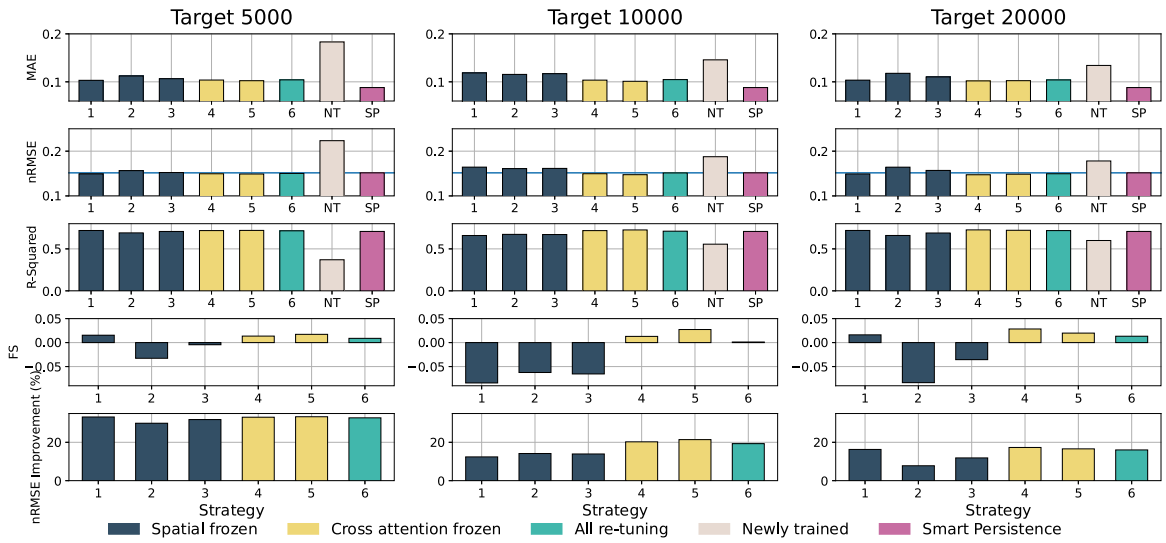


Fig. 10. Comparison between metrics after the transfer learning to the target domain. (NT denotes newly trained models without the pre-trained parameters, the nRMSE improvement row represents the percentage enhancement of transferred models to the NT models, and the FS row refers to (11) on the target dataset.).

Table 3
Comparison of with and without temporal block.

Model	MAE	nRMSE	R-Square	FS
DSCAT	0.050	0.093	0.870	0.490
Without Temporal Block	0.055	0.102	0.842	0.437
Relative Improvement	8.32%	9.31%	3.33%	**11.97%**

Additionally, an ablation study was conducted with and without integrating the temporal block to prove the function of the double route and cross-attention temporal structure. The performance metrics are shown in Table 3. In this table, it is clear that across all the metrics, the DSCAT outperforms the model without integrating the double route and cross-attention temporal structure, the FS value improvement by integrating this structure reaches 11.97%. Moreover, the spatio-temporal query-key pair mechanism provides a better correlation coefficient score between the input and output.

Finally, the sizes of all the models are compared in Fig. 9. It shows that the Transformer-based models have considerably smaller sizes than the CNN-based structures. Although the size of the DSCAT is slightly larger than that of the ViT, it is less than 100 MB to provide convenient transferability.

5.2. Transfer performance on target domain

The above PV forecasting results on the source domain show that the DSCAT outperforms other deep learning models on an information-sufficient database. After that, the transfer strategies are implemented on various lengths of the target dataset to simulate the model performance on newly built PV sites. The simulated database length (number of data points) is set as 5000, 10000, and 20000, corresponding to nearly one week, two weeks, and four weeks, respectively.

The metrics are shown in Fig. 10, the NT denotes the newly trained models without the pre-trained parameters. Each column in this figure denotes a different size of the target dataset, and each row shows a distinct metric. The nRMSE improvement row represents the percentage enhancement of transferred models to the NT models. The FS is compared based on the nRMSE of the smart persistence algorithm on the target test set. From left graphs to right graphs, the size of the target dataset is increased. It can be seen that the models with pre-trained parameters perform better than the NT model across all the metrics and with varied sizes of target datasets. However, only the strategies 4, 5, and 6 performed better than the smart persistence algorithm. Moreover, in six transfer strategies (refer to Fig. 5), strategies 4 and 5 show outstanding performance than the others in all metrics. Refer to Fig. 5, strategies 4 and 5 only have the deep spatial block and spatio-temporal cross-attention block frozen. It proves that the largest difference between the source and target PV datasets is the spatial feature distribution of the sky images. After the extraction of dissimilar spatial features by the shallow spatial layers, the patterns of the essential hidden features are common between the source and target domains, which can be analyzed by the same spatio-temporal prediction block. With the increasing dataset size, the average FS scores of strategies 4, 5, and 6 increase. This implies that the transferred DSCAT model can study and grow with more data, while the smart persistence algorithm could only provide fixed performance. Overall, Fig. 10 proves that the knowledge transferred from the source domain to the target domain can improve the model performance, and the PV forecasting for the newly built PV sites could be enhanced by the DSCAT framework which is pre-trained on the large PV dataset.

Further, to evaluate the performance of the DSCAT framework, the benchmark models are all transferred to the target dataset (size of 10000). As most of the benchmark models do not have distinct modular structures, three transfer strategies are implemented on the benchmarks respectively: (A) Freezing all fully connected layers, which are the output layers of all benchmarks; (B) Freezing the fully connected layers and all LSTM-based layers, which are the deep layers like the strategies 4 and 5 in Fig. 5; and (C) Fine-tuning on all layers. The transferring results and comparisons are shown in Table 4. In Table 4, NT also denotes the newly trained model on the target dataset, and the FS(DSCAT) uses the nRMSE of DSCAT as the standard to calculate the FS score on the transferred benchmarks. This table shows that even though some of the strategies on the benchmark models could provide smaller nRMSE than the NT versions, none of the transferred benchmarks can have a positive FS score against the smart persistence algorithm. That means all the transfer strategies of benchmarks are not applicable to the real PV forecasting transferring task. In comparison to the transferred DSCAT with strategy 5, the best performance comes from the ConvLSTM with benchmark strategy C. However, the best transfer benchmark still has a 26% gap with respect to the transferred DSCAT model. In the comparisons, it could be seen that with the distinct modular structure and better capability of feature extraction, the DSCAT has an outperforming performance in transferring to the newly built PV sites.

Table 4
Metrics of transferred benchmarks on target dataset (size 10000).

Strategy	Model	MAE	nRMSE	R-Square	FS	FS(DSCAT)
A	CNN	0.150	0.202	0.487	−0.330	−0.372
	CNNLSTM	0.144	0.190	0.545	−0.253	−0.292
	ConvLSTM	0.164	0.213	0.428	−0.405	−0.449
	ViT	0.163	0.217	0.407	−0.430	−0.474
B	CNNLSTM	0.155	0.196	0.516	−0.292	−0.332
	ConvLSTM	0.154	0.210	0.443	−0.386	−0.429
C	CNN	0.142	0.188	0.555	−0.239	−0.278
	CNNLSTM	0.140	0.189	0.551	−0.244	−0.283
	ConvLSTM	0.143	0.185	0.568	−0.222	**−0.260**
	ViT	0.161	0.215	0.415	−0.421	−0.465
NT	CNN	0.146	0.195	0.523	−0.283	−0.323
	CNNLSTM	0.153	0.207	0.458	−0.368	−0.411
	ConvLSTM	0.142	0.190	0.544	−0.254	−0.293
	ViT	0.157	0.211	0.440	−0.390	−0.434
	SP	**0.088**	0.152	0.708	N/A	N/A
	DSCAT (Strategy 5)	0.101	**0.147**	**0.726**	**0.027**	N/A

N/A: Not the target to evaluate

6. Conclusion

This paper proposed an effective, transferable, and readily deployable ultra-short-term PV forecasting framework DSCAT. The shifted-window Transformer structure and the double route cross-attention mechanism are combined in this framework. It can effectively capture the spatial and temporal features embedded in the sky image series. The DSCAT achieves a 49% improvement over the smart persistence baseline and an average 13% improvement over several representative deep learning-based PV forecasting benchmarks. This paper also investigates the transferability of various forecasting model structures. It shows that after the model transferring between different datasets, unlike the deep learning-based PV forecasting benchmarks, which underperformed compared to the smart persistence baseline, the DSCAT can transfer the knowledge of weather patterns acquired from the established PV site (having a large size of historic PV data) to the newly constructed PV site (having limited PV data). The performance of the transferred model could be further improved with the increase in data from the target PV site.

The proposed DSCAT framework provides a high-accuracy PV generation forecasting structure and demonstrates the potential of using sky images to enhance PV generation forecasting on newly constructed PV sites. In future work, other data sources from the newly constructed PV sites such as satellite images and weather data will also be integrated.

CRediT authorship contribution statement

Shijie Xu: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Hui Ma:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Chandima Ekanayake:** Writing – review & editing, Supervision, Conceptualization. **Yi Cui:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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